Instrumental Variables

Session 1: Foundations and Mechanics

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Empirical Methods for PhD Students

Session 1 Roadmap

Outline

Three Conditions for a Valid Instrument

A valid instrument Z_i must satisfy three conditions:

1. **Relevance (First Stage):** Z_i must be correlated with the endogenous treatment D_i
   - Testable: first-stage F-statistic (rule of thumb: $F > 10$)
2. **Independence (As-Good-As-Random Assignment):** Z_i must be uncorrelated with unobserved confounders ε_i
   - Not directly testable; requires economic reasoning and robustness checks
3. **Exclusion Restriction:** Z_i affects outcome Y_i *only* through its effect on D_i
   - No direct effect; testable indirectly via placebo/falsification

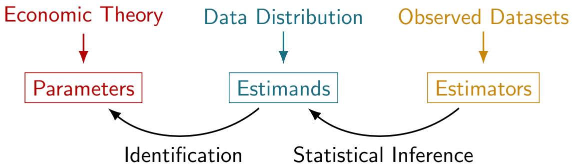
In Session 2 we add a 4th condition: **Monotonicity** (for the LATE interpretation).

Parameters, Estimands, and Estimators

Three distinct objects, not always clearly distinguished:

- Parameters come from economic (or other) models of the world
  - E.g. a potential-outcome model relating schooling to earnings
  - They set the *target* for empirical analyses: what we want to know
- Estimands are functions of the population data distribution
  - E.g. a difference in means or ratio of population regression coefficients
  - Identification: assumptions that link parameters to estimands
- Estimators are functions of observed (sample) data
  - E.g. a sample ratio of OLS coefficients
  - Estimators are random variables; statistical inference studies their distribution

The Lay of the Land



Separating identification (linking parameters to estimands) from inference (linking estimators to estimands) keeps the analysis disciplined.

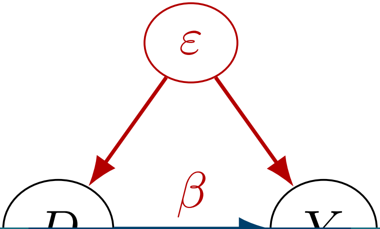
Simple causal model: $Y_i = \beta D_i + \varepsilon_i$, where D_i is the treatment, ε_i is the unobserved term.

In large samples, OLS converges to:

$$\beta^{OLS} = \beta + \frac{Cov(\varepsilon_i, D_i)}{Var(D_i)}$$

- OLS identifies β *only if* $Cov(\varepsilon_i, D_i) = 0$
- **Selection bias**: units with higher/lower ε_i select into treatment, so $Cov(\varepsilon_i, D_i) \neq 0$

Causal diagram (endogeneity):



The IV Solution

Suppose $Z_i \in \{0,1\}$ is a binary instrument (e.g. lottery assignment).

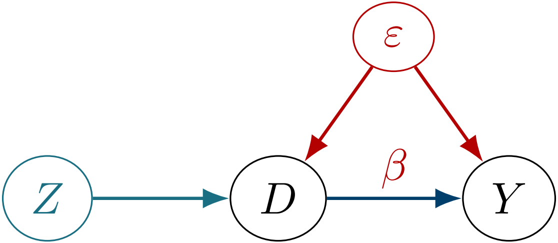
- **Randomness**: $Cov(Z_i, \varepsilon_i) = 0$ (no direct link to unobservables)
- **Exclusion**: Z_i affects Y_i only through D_i

Plugging in $\varepsilon_i = Y_i - \beta D_i$:

$$Cov(Z_i, Y_i - \beta D_i) = 0 \implies \beta = \frac{Cov(Z_i, Y_i)}{Cov(Z_i, D_i)} \equiv \beta^{IV},$$

so long as $Cov(Z_i, D_i) \neq 0$ (relevance).

Causal diagram (IV adds Z):



No arrow from Z to ε (independence) and no arrow from Z to Y

Reduced Form and First Stage

A simple decomposition links IV back to familiar OLS regressions:

$$\beta^{IV} = \frac{Cov(Z_i, Y_i)}{Cov(Z_i, D_i)} = \frac{Cov(Z_i, Y_i)/Var(Z_i)}{Cov(Z_i, D_i)/Var(Z_i)} \equiv \frac{\rho}{\pi}$$

where ρ and π come from two OLS regressions:

$$\begin{aligned} Y_i &= \kappa + \rho\, Z_i + \nu_i \quad (\text{“reduced form”}) \\ D_i &= \mu + \pi\, Z_i + \eta_i \quad (\text{“first stage”}) \end{aligned}$$

- The **first stage** shows how strongly Z_i moves D_i
- The **reduced form** shows the net effect of Z_i on Y_i
- IV scales the reduced form by the first stage to recover β

Adding Controls

If Z_i is as-good-as-random only *conditional* on covariates W_i, add them to both equations:

$$Y_i = \kappa + \rho Z_i + W_i'\phi + \nu_i$$
$$D_i = \mu + \pi Z_i + W_i'\psi + \eta_i$$

- The Frisch-Waugh-Lovell theorem tells us the effective instrument is now $\tilde{Z}_i$: the residuals from regressing Z_i on W_i
- If W_i is a set of randomization-strata dummies, $\tilde{Z}_i$ captures within-strata variation in Z_i
- Key: controls should be chosen on substantive grounds, not to maximize the first-stage F-statistic

Classic Example: Angrist (1990) Vietnam Draft Lottery

Angrist (1990, AER) estimated the earnings effect of military service using Vietnam-era draft-eligibility as an instrument.

- $Z_i \in \{0, 1\}$: randomly assigned draft eligibility (lottery number)
- $D_i \in \{0, 1\}$: actual military service
- Y_i: later-life earnings

With binary Z_i, the IV estimand simplifies to the **Wald estimand**:

$$\beta^{IV} = \frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]}$$

- Numerator: earnings effect of draft eligibility (reduced form)
- Denominator: effect of eligibility on actual service probability (first stage)
- Finding: military service reduces early-career earnings by $\approx 15\%$

Outline

Multiple Instruments: Overidentification

When $\dim(Z_i) = L > J = \dim(X_i)$, the system is *overidentified*:

$$Cov(\tilde{Z}_i, Y_i - X_i'\beta) = 0 \implies \underbrace{Cov(\tilde{Z}_i, Y_i)}_{L\times 1} = \underbrace{Cov(\tilde{Z}_i, X_i')}_{L\times J}\underbrace{\beta}_{J\times 1}$$

- We have more moment conditions than parameters
- Any full-rank $J \times L$ matrix M gives a valid IV estimator: $\beta^{IV}(M) = \left(M\, Cov(\tilde{Z}_i, X_i')\right)^{-1} M\, Cov(\tilde{Z}_i, Y_i)$
- Choice of M trades off efficiency vs. interpretability

Two-Stage Least Squares (2SLS)

2SLS chooses $M' = \pi$ (the matrix of first-stage coefficients):

- Effective instrument: $\tilde{Z}_i^* = \tilde{Z}_i'\pi$ (residualized first-stage fitted values)
- Combine instruments according to their predictive power for $X_i \Rightarrow$ minimizes the 2SLS standard error

Two-stage interpretation:

1. *First stage*: regress X_i on Z_i and W_i; save fitted values $\hat{X}_i$
2. *Second stage*: regress Y_i on $\hat{X}_i$ and W_i

Practical warning

Never run 2SLS by hand—the second-stage standard errors are wrong. Use `ivreg2` (Stata) or `AER::ivreg()` / `fixest::feols()` (R).

2SLS as a Weighted Average of Just-Identified IVs

When there is a single treatment ($J = 1$), 2SLS has an elegant weighted-average representation:

$$\beta^{2SLS} = \sum_{\ell} \omega_\ell \, \beta_\ell^{IV}, \quad \omega_\ell = \frac{\pi_\ell^2 Var(\tilde{Z}_{i\ell})}{\sum_{\ell'} \pi_{\ell'}^2 Var(\tilde{Z}_{i\ell'})}$$

- $\beta_\ell^{IV} = \rho_\ell / \pi_\ell$ is the just-identified IV estimate using instrument ℓ alone
- Weights ω_ℓ are proportional to each instrument's contribution to first-stage variation
- Instruments with stronger first stages receive larger weights
- This "visual IV" representation (slope of ρ_ℓ on π_ℓ) is also useful for diagnosis

Overidentification Tests

Under the *constant-effects* model, all just-identified IVs identify the same β:

- If β_ℓ^{IV} differ significantly across instruments, something is wrong
- The Sargan–Hansen statistic tests $H_0 : \beta_\ell^{IV} = \beta$ for all ℓ; asymptotically $\chi^2(L - J)$ under the null

Cautions:

- Low power when individual $\hat{\beta}_\ell^{IV}$ are noisy
- Rejection does *not* tell you which instrument is invalid
- Under *heterogeneous* effects (Session 2), two valid IVs can legitimately have different estimands; the test is not interpretable
- Use the test as one input among many, not as a pass/fail criterion

Outline

Weak Instruments

If π is small relative to its standard error, the instrument is "weak":

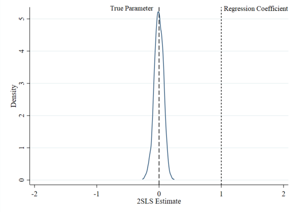
- Rule of thumb: first-stage $F < 10$ signals potential weakness (Staiger and Stock, 1997)
- **Bias**: 2SLS is biased toward OLS in small samples; the bias is proportional to $1/F$ approximately
- **Size distortion**: standard confidence intervals may under-cover in finite samples

Modern guidance (Angrist and Kolesár 2022):

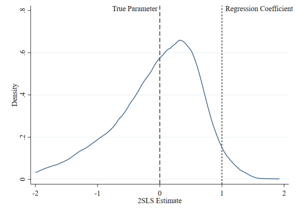
- In the just-identified case, 2SLS is *median-unbiased*; the SE increase tends to "cover up" the bias
- Use `weakivtest` (Stata; Montiel Olea and Pflueger 2015) for more nuanced critical values
- If F is small: consider Anderson–Rubin or JIVE as robust alternatives

$Y_i = 0 \cdot D_i + \varepsilon_i$, $D_i = \pi Z_i + \eta_i$, $N = 100$ replications

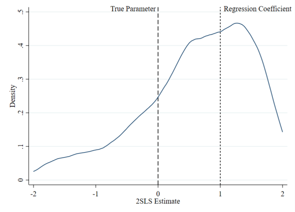
Strong ($\pi = 1$, $F \approx 100$)



Medium ($\pi = 0.1$, $F \approx 10$)



Weak ($\pi = 0.01$, $F \approx 1$)



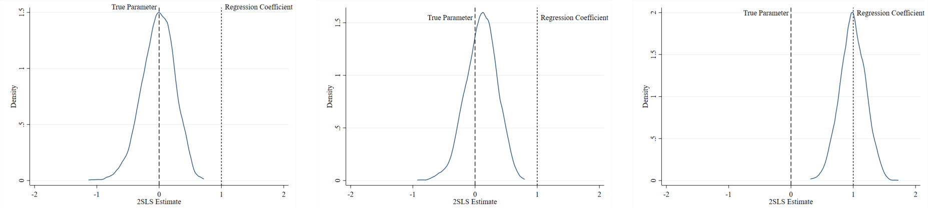
Distribution of $\hat{\beta}^{IV}$ widens dramatically as the instrument weakens. With a very weak instrument, $\hat{\beta}^{IV}$ converges to $\hat{\beta}^{OLS}$.

Overidentified 2SLS faces an additional problem when there are many weak instruments:

- **Symptom**: low first-stage F even though SEs are small (more IVs $\Rightarrow$ richer first stage $\Rightarrow$ better fit of D_i)
- **Mechanism**: overfitting the first stage essentially recreates the endogenous variation in D_i, biasing 2SLS toward OLS
- **Historical case**: Angrist–Krueger (1991) QoB IV interacted with state $\times$ year fixed effects created hundreds of instruments

Visualized — 100 replications:

Few IVs | 10 IVs | 100 IVs



Practical Guidance on Instrument Strength

Prevention:

- Prefer parsimonious instrument sets; avoid mechanical many-IV settings
- If using a Bartik/shift-share IV with many industries, the *effective* number of instruments is the number of industries (addressed in Session 2)

Diagnosis:

- Report first-stage F-statistic (or Kleibergen–Paap for clustered SEs)
- Use `weakivtest` in Stata for heteroskedasticity-robust critical values
- See also Lee et al. (2020) for t-ratio-based inference

If the instrument is weak:

- Try a more informative functional form for Z_i
- Collapse to fewer instruments
- Use Anderson–Rubin confidence sets (valid regardless of strength)

Outline

Session 1 Applications: Four Classic IV Papers

We study four highly cited papers illustrating core IV designs:

Decision-maker leniency IV (Western context):

- Maestas, Mullen, and Strand (2013, AER)
  "Does Disability Insurance Receipt Discourage Work?"
  IV: quasi-randomly assigned SSDI disability examiners.
- Aizer and Doyle (2015, QJE)
  "Juvenile Incarceration, Human Capital, and Future Crime"
  IV: randomly assigned juvenile court judges.

Geographic and historical IV (Chinese context):

- Qian (2008, QJE)
  "Missing Women and the Price of Tea in China"
  IV: county agricultural suitability for tea cultivation.
- Chen, Kung, and Ma (2020, QJE)
  "Long Live Keju!"
  IV: historical imperial examination quota allocation.

Paper 1: Maestas, Mullen, and Strand (2013, AER)

Citation

Maestas, N., Mullen, K.J., and Strand, A. (2013). "Does Disability Insurance Receipt Discourage Work? Using Examiner Assignment to Estimate Causal Effects of SSDI Receipt." *American Economic Review*, 103(5): 1797–1829. (≈1,300 Google Scholar citations)

Research question: Does receiving Social Security Disability Insurance (SSDI) benefits *cause* a reduction in employment?

Context & motivation:

- SSDI provides income support to workers with disabling conditions; by 2010 over 8.2 million Americans received benefits ($119 billion)
- Key policy concern: does SSDI create a "poverty trap" by discouraging recipients from returning to work?
- OLS is severely biased: healthier rejected applicants would work anyway; sicker accepted applicants could not work regardless (*reverse causality* + *selection on unobserved health*)

MMS (2013): Examiner Assignment IV Strategy

The IV: Randomly assigned disability examiners differ systematically in their allowance rates ("leniency"):

- SSDI applications are assigned to examiners quasi-randomly within local Disability Determination Services (DDS) offices
- Examiners vary substantially in their allowance rates even after controlling for case characteristics
- An examiner's average grant rate (leave-one-out, in other cases) is used as the instrument for the applicant's SSDI receipt

IV validity:

- Relevance: examiner leniency strongly predicts own award probability (first-stage $F > 100$)
- Independence: assignment is quasi-random within offices (applicants cannot choose their examiner)
- Exclusion: examiner leniency affects employment only through SSDI award, not through other channels
- Monotonicity: a stricter examiner can only reduce (never increase) an

MMS (2013): Data and Specification

Data:

- Social Security Administration administrative records: ≈1.5 million SSDI applicants, 1997–2004
- **Treatment**: binary indicator for SSDI award (0/1)
- **Outcomes**: employment indicator, earnings (linked to earnings records from IRS W-2 forms)
- **Instrument**: leave-one-out examiner allowance rate (average grant rate of the assigned examiner on all other cases)
- **Strata**: DDS office × quarter fixed effects (controls for within-office assignment patterns)

Specification:

$$Y_{it} = \alpha + \beta\,\mathsf{SSDI}_i + W_i'\gamma + \varepsilon_i$$

- 2SLS using examiner leniency as Z_i, conditional on DDS-quarter FEs
- Robust to controlling for applicant characteristics (age, diagnosis, prior earnings) because assignment is as good as random

MMS (2013): Results and Robustness

Key findings:

- SSDI receipt reduces employment by ≈28 percentage points at ages 45–64 (IV estimate much larger than OLS)
- About 23% of marginal SSDI recipients (compliers) could work: the "work-capable" group whose decision is affected by examiner assignment
- OLS understates the causal effect because rejected applicants who remain employed create a spurious positive "health effect"

Robustness and sensitivity:

- Placebo test: examiner leniency does not predict pre-application employment trends (balance on pre-period outcomes)
- Alternative examiner measures: using quintiles of leniency instead of the continuous measure gives nearly identical results
- Sub-group consistency: effects are similar across age groups, diagnosis categories, and states
- Complier analysis: marginal recipients are younger, healthier, and have higher prior earnings than average recipients, explaining why SSDI receipt

MMS (2013): Contribution and IV Lessons

Academic contributions:

- First credible causal estimate of SSDI's effect on employment; resolves a long-standing identification debate
- Shows SSDI creates work disincentives for a sizeable share of recipients—important for disability insurance design
- Introduced the "examiner leniency" IV into economics; spawned a large literature on disability insurance, screening, and physician/administrative judgment

IV methodology lessons:

- Random assignment of decision-makers is a powerful source of exogenous variation when direct randomization is infeasible
- The leave-one-out leniency measure avoids mechanical correlation between one's own case and the instrument
- LATE interpretation is transparent: IV estimates the effect of SSDI on "compliers"—those whose award decision is influenced by examiner assignment

Paper 2: Aizer and Doyle (2015, QJE)

Citation

Aizer, A. and Doyle, J.J., Jr. (2015). "Juvenile Incarceration, Human Capital, and Future Crime: Evidence from Randomly Assigned Judges." *Quarterly Journal of Economics*, 130(2): 759–803. (≈900 Google Scholar citations)

Research question: Does incarcerating juveniles reduce or increase their subsequent criminal activity and human capital?

Context & motivation:

- The US incarcerates approximately 70,000 juveniles on any given day; recidivism rates are high (≈70% re-arrested within 3 years)
- Policy debate: does incarceration deter future crime (deterrence), or does it increase it via criminal networks and disrupted schooling?
- OLS is biased: more serious offenders are both more likely to be incarcerated *and* more likely to re-offend regardless

Aizer & Doyle (2015): Judge IV Strategy

The IV: Randomly assigned juvenile court judges vary in their propensity to incarcerate:

- Juvenile cases are assigned to judges randomly within courthouses in Chicago, Illinois
- Judges differ substantially in their incarceration rates even for similar offense types and defendant characteristics
- A judge's leave-one-out incarceration rate on other cases is the instrument for whether defendant i is incarcerated

IV validity:

- Relevance: judge leniency strongly predicts incarceration (first-stage $F \approx 35$–50)
- Independence: assignment is explicitly random within court (confirmed by balance tests on pre-determined characteristics)
- Exclusion: judge assignment affects outcomes only through the incarceration decision, not through other channels
- Monotonicity: a stricter judge cannot reduce a defendant's incarceration

Aizer & Doyle (2015): Data and Specification

Data:

- Illinois Juvenile Court data linked to adult criminal records, high school graduation records, and mortality data; 1990–2006
- ≈35,000 juvenile cases in Chicago (ages 10–17)
- **Treatment**: binary incarceration indicator (0/1)
- **Outcomes**: adult incarceration, high school graduation, employment, and earnings in adulthood (ages 25–35)
- **Instrument**: leave-one-out judge incarceration rate (within courthouse, controlling for offense type and year)

Specification:

$$Y_i = \alpha + \beta \, \text{Incarcerated}_i + W_i' \gamma + \varepsilon_i$$

- 2SLS using judge leniency as Z_i, with courthouse-year-offense category fixed effects as strata
- Outcomes measured 5–15 years after the juvenile offense

Aizer & Doyle (2015): Results and Robustness

Key findings:

- Juvenile incarceration reduces high school graduation by ≈13 percentage points
- Juvenile incarceration increases adult incarceration probability by ≈22–26 percentage points
- Earnings are reduced significantly; effects persist for decades
- Net cost-benefit analysis: incarceration costs society roughly $1.1–1.4 million per juvenile (present value), far exceeding any deterrence benefit

Robustness and sensitivity:

- Balance test: judge leniency is uncorrelated with defendant race, prior offenses, and offense severity (random assignment holds)
- Placebo: judge leniency does not predict pre-offense outcomes (no pre-existing differences across complier groups)
- Different charge types: results hold across assault, drug, property, and public order offenses
- Bandwidth variation: narrowing the leave-one-out window (same offense

Aizer & Doyle (2015): Contribution and IV Lessons

Academic contributions:

- First causal evidence that juvenile incarceration increases adult criminality and reduces human capital—a critical finding for criminal justice reform debates
- Demonstrates that the criminal justice system can perpetuate rather than reduce criminal behavior through incapacitation effects
- Extended the judge-leniency IV approach (Kling 2006) to juvenile settings and linked to long-run human capital outcomes

IV methodology lessons:

- Judge/examiner leniency IV is now a standard tool whenever administrative decisions are quasi-randomly assigned
- Key to validity: random (not quasi-random) assignment within strata is what makes the exclusion credible
- LATE: IV identifies the effect among "compliers"—juveniles whose incarceration decision is swayed by judge assignment, typically those near the incarceration margin

Comparing the Two Papers

	MMS (2013)	**AD (2015)**
Journal	AER	QJE
Setting	US SSDI applicants	US juvenile defendants
Treatment	SSDI benefit receipt	Juvenile incarceration
IV type	Examiner leniency	Judge leniency
Assignment	Quasi-random (within DDS office)	Random (within courthouse)
Unit	Individual applicant	Individual defendant
Key finding	SSDI $\downarrow$ employment 28pp	Incarceration $\downarrow$ graduation 13p
Citations	$\approx$1,300	$\approx$900

Common thread: Both papers exploit **random variation in decision-maker leniency** to identify causal effects of high-stakes policy interventions. The IV design is clean, credible, and transparent — a template for policy evaluation research.

Citation

Qian, N. (2008). "Missing Women and the Price of Tea in China." *Quarterly Journal of Economics*, 123(3): 1251–1285. (≈3,500 Google Scholar citations)

Research question: Does the relative economic value of female labor reduce female mortality (the "missing women" phenomenon)?

Context & motivation:

- Amartya Sen (1990): ≈100 million "missing" women in Asia due to son preference, sex-selective abortion, and differential care
- China's 1978–79 decollectivization let farmers sell surplus produce; crop suitability became critical for household income
- OLS is biased: regions with higher female wages may differ in institutions, culture, and law enforcement

Qian (2008): Geographic IV Design

The IV: County-level agricultural suitability for tea vs. orchard cultivation (determined by soil acidity, rainfall, and elevation):

- Tea cultivation is female-labor intensive; orchards are male-labor intensive—so tea suitability raises relative female wages
- Suitability is geographically predetermined and exogenous to son-preference norms and government policy
- Treatment: post-reform relative female income; Outcome: sex ratio at ages 0–5 and girls' survival rates

IV validity:

- Relevance: tea suitability $\times$ post-1978 period strongly predicts female-to-male income ratio (first-stage $F > 40$)
- Independence: soil/terrain characteristics are not correlated with gender preferences or other cultural confounders
- Exclusion: tea suitability affects sex ratios only through its effect on relative female income (falsification: no pre-reform relationship between tea suitability and sex ratios)

Qian (2008): Results and IV Lessons

Key findings:

- A 10% increase in female relative income reduces the sex ratio at ages 0–5 by ≈2–3%—significant recovery of missing girls
- Effects are driven by increased female survival, not changes in sex-selective abortion—consistent with reduced neglect and better care
- Results survive controls for pre-existing trends, income levels, and province fixed effects

IV methodology lessons:

- Geographic/agricultural suitability (soil, terrain, climate) is a classic source of exogenous variation in development economics— it directly affects economic activity but is predetermined with respect to social outcomes
- The pre/post-reform DID structure is transparent: tea-suitable counties gain more after 1978, creating cross-sectional IV variation
- Falsification: no relationship between tea suitability and pre-reform sex ratios confirms the independence assumption

Paper 4: Chen, Kung, and Ma (2020, QJE) — Long Live Keju!

Citation

Chen, T., Kung, J.K.S., and Ma, C. (2020). "Long Live Keju! The Persistent Effects of China's Imperial Examination System." *Quarterly Journal of Economics*, 135(4): 1801–1846. (≈600 Google Scholar citations)

Research question: Did China's Imperial Examination System (Keju, 605–1905 AD) leave persistent effects on human capital and development?

Context & motivation:

- The Keju selected government officials through standardized exams; meritocracy incentivized education across the population
- Counties differed dramatically in exam-slot quotas due to administrative allocation—creating cross-county variation
- OLS is biased: counties with more scholars may have had better institutions, more resources, or geographic advantages

Chen, Kung & Ma (2020): Historical Quota IV

The IV: Imperial-era Keju examination quotas allocated to each county by the Qing government based on tax revenue:

- Quotas were assigned for fiscal/administrative convenience—not educational merit—creating plausibly exogenous variation
- Higher quotas ⇒ more aspiring candidates ⇒ stronger culture of human capital investment
- Treatment: historical Keju scholar density per county; Outcomes: modern GDP, education levels, private entrepreneurship

IV validity:

- Relevance: quota allocation strongly predicts scholar density (first-stage $F > 50$)
- Independence: quotas uncorrelated with modern economic potential after controlling for the county tax base (the source of quota variation)
- Exclusion: quotas affect modern outcomes only through cultural/human capital persistence; direct fiscal effects from Qing-era quotas have long dissipated

Chen, Kung & Ma (2020): Results and IV Lessons

Key findings:

- A 1-SD increase in Keju scholar density raises modern county GDP per capita by ≈6–8%
- Effects operate through education channels: higher Keju exposure ⇒ more schools, higher literacy, more private entrepreneurs
- Robust to controls for geography, Ming-dynasty institutions, province fixed effects, and multiple outcome time periods

IV methodology lessons:

- Historical administrative records provide rich sources of exogenous variation when modern experimental designs are infeasible
- Long-run persistence studies rely heavily on a credible exclusion restriction: document and test all plausible alternative channels
- The exclusion is bolstered by mediation analysis: tracing the Keju → education → development pathway directly

Outline

Session 1 Summary

IV Foundations:

- IV validity requires relevance, independence, and exclusion
- $\beta^{IV} = \rho/\pi$ (reduced form $\div$ first stage)
- With controls: effective instrument $\tilde{Z}_i$ = residual of Z_i on W_i (Frisch–Waugh)

2SLS and Overidentification:

- 2SLS = efficiency-optimal combination of multiple instruments
- 2SLS = weighted average of just-identified IVs
- Sargan–Hansen test: diagnostic, not a "certification of validity"

Instrument Strength:

- $F > 10$ rule of thumb; `weakivtest` for robust critical values
- Many-IV bias: overfitting first stage recreates endogeneity

Applications (Top-5 Journals):

- MMS (2013, AER): examiner leniency IV $\to$ SSDI reduces employment by

Preview: Session 2

1. **IV Interpretation (LATE)**
   - Imbens–Angrist (1994) assumptions: monotonicity, compliers
   - Characterizing compliers; diff-in-diff IV
2. **Formula Instruments**
   - Shift-share IV: formal conditions (Borusyak et al. 2022)
   - Recentered IV: beyond shift-share (Borusyak and Hull 2023)
   - Illustration: China High-Speed Rail market access
3. **Chinese Economic Applications**
   - Meng, Qian, Yared (2015, *ReStud*): Great Famine IV
   - Khandelwal, Schott, Wei (2013, *QJE*): WTO quota removal

Instrumental Variables — Session 1

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Questions welcome via email.

Session 2 slides cover IV interpretation and Chinese applications.